Predictive and Generative AI

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Predictive and Generative AI before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone Expo

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Predictive and Generative AI trace and record the first step you can explain without help.

Use W2: Compute the distribution for Generative and predictive objectives. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Predictive and Generative AI explanation against the supplied facts before discussing it.

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each **mechanism** and the **evidence** needed to check its **output**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your learning target

> Connect the Predictive and Generative AI target to a trace and a justified conclusion.

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Predictive and Generative AI case and show the deciding evidence.

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 1 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Predictive and Generative AI adds to Verified Study Buddy and keep the evidence.

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Identify the objective

> Identify the objective. Apply the supplied Predictive and Generative AI facts and keep your reasoning.

Classify these components by **target**: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compute the distribution

> Compute the distribution. Apply the supplied Predictive and Generative AI facts and keep your reasoning.

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Follow the prefix

> Follow the prefix. Apply the supplied Predictive and Generative AI facts and keep your reasoning.

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Analyse a new corpus independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Analyse a new corpus independently. Apply the supplied Predictive and Generative AI facts and keep your reasoning.

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a useful evaluation

> Design a useful evaluation. Apply the supplied Predictive and Generative AI facts and keep your reasoning.

A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

> **My reasoning and evidence**
>
> _____
>
> _____
>
> _____
>
> _____

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

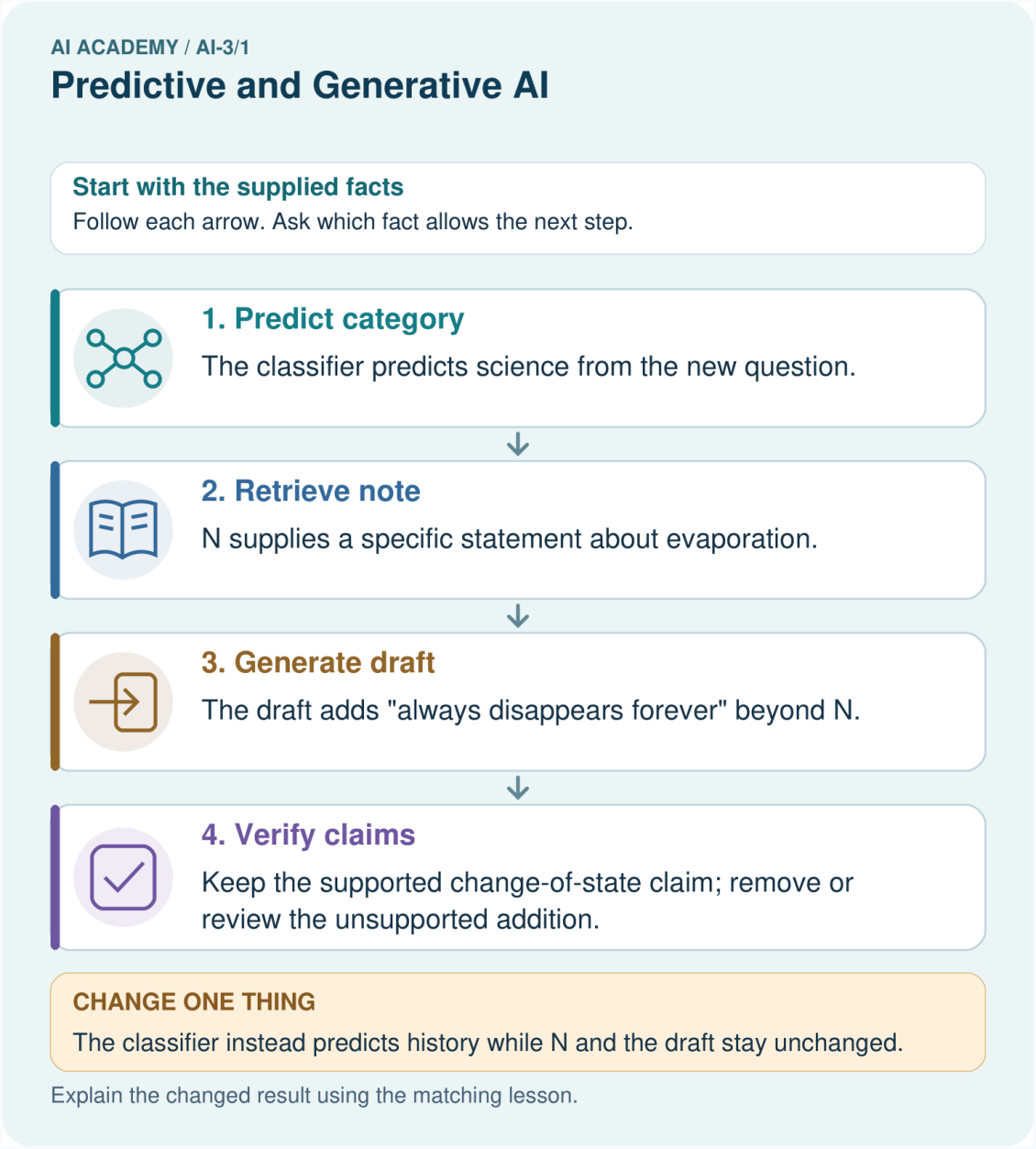
Pictorial case practice: Predictive and Generative AI

> A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Explain the labelled step before checking the changed condition.

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”



Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the first unsupported claim and explain what source would be needed before asserting it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep the supported change-of-state claim; remove or review the unsupported addition.

1. Predict category

2. Verify claims

3. Retrieve note

4. Generate draft

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Verify claims Keep the supported change-of-state claim; remove or review the unsupported addition. Classification predicts a category under a trained mapping.

Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The classifier instead predicts history while N and the draft stay unchanged. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The category **prediction** changes, but the same unsupported draft claim still needs checking.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The category **prediction** changes, but the same unsupported draft claim still needs checking. Stages can fail independently; one result does not validate another.

Repair a missing building block

Review Capstone Expo (ai-2/36). Goal: Defend an application using reproducible **evidence**. Short practice: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

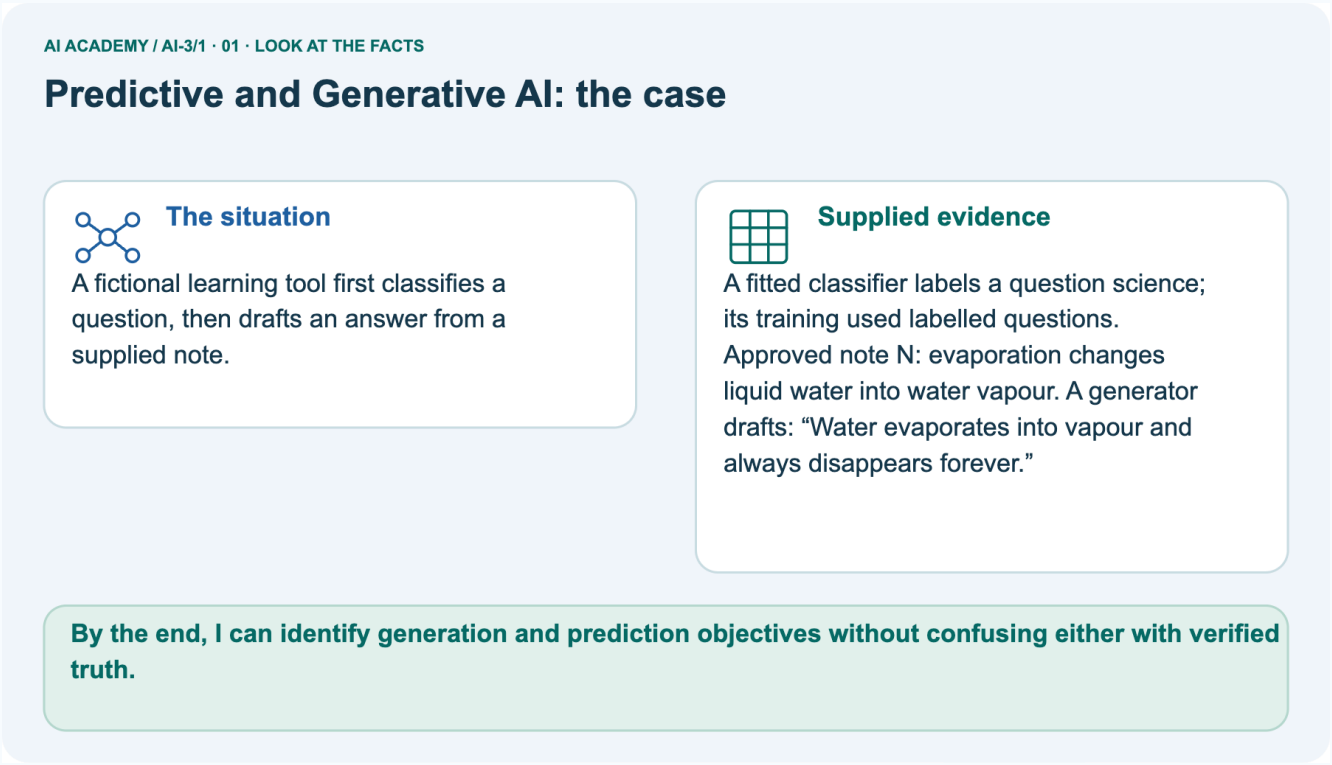
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Predictive and Generative AI picture from its labels, then record a relevant limit.

01. Predictive and Generative AI: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

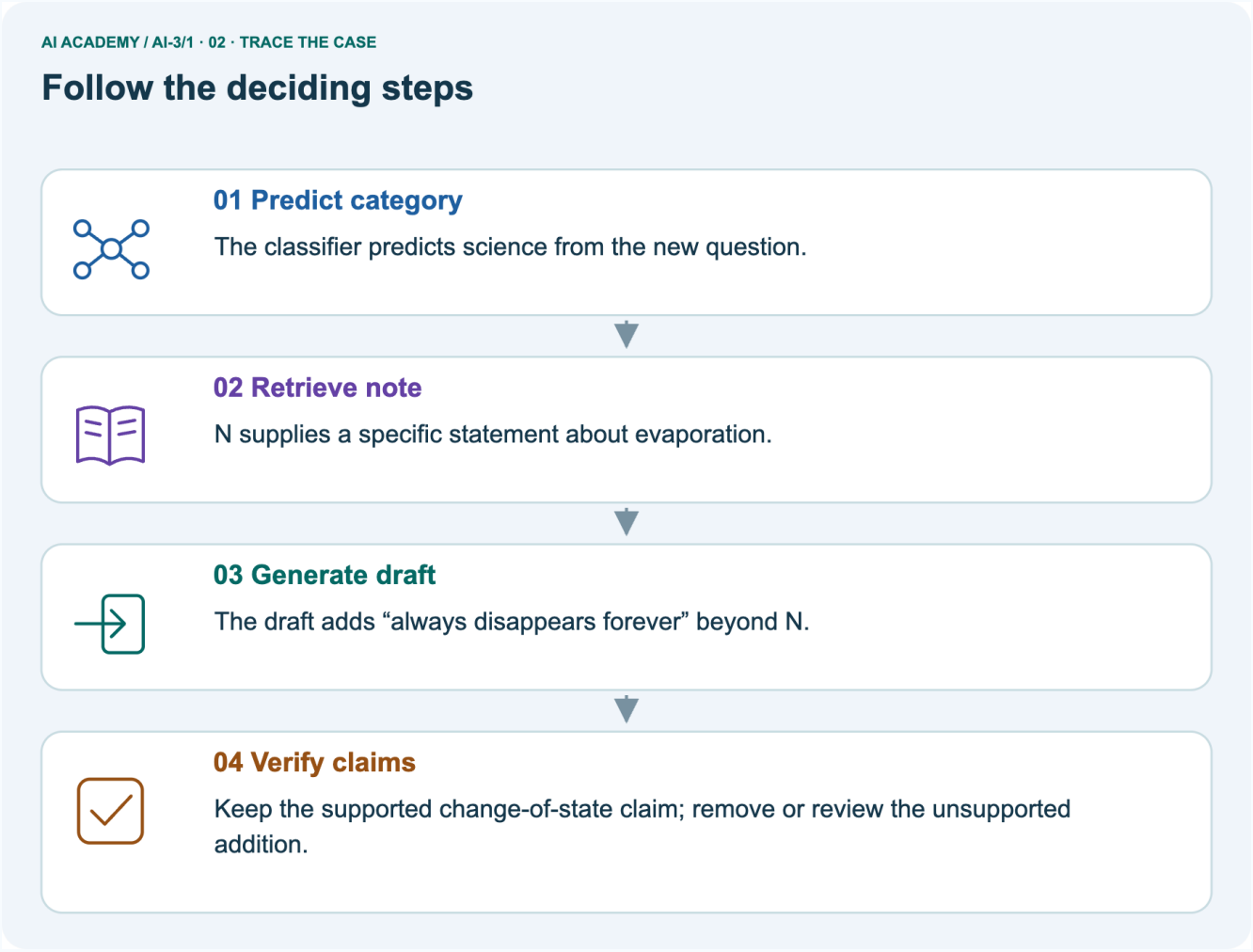
A fictional learning tool first classifies a question, then drafts an answer from a supplied note. Supplied facts: A fitted classifier labels a question science; its **training** used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



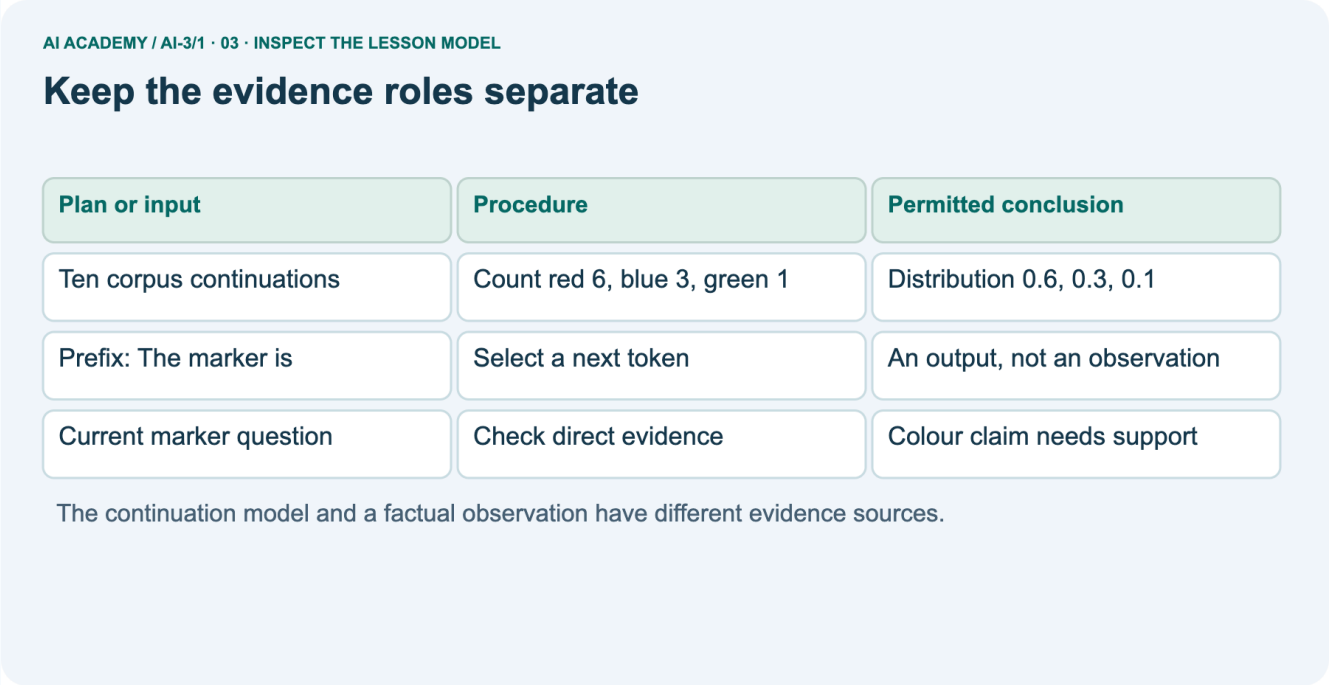
Follow the labelled picture with the matching explanation. Open the vector diagram

Predict category: The classifier predicts science from the new question. → Retrieve note: N supplies a specific statement about evaporation. → Generate draft: The draft adds “always disappears forever” beyond N. → Verify claims: Keep the supported change-of-state claim; remove or review the unsupported addition.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

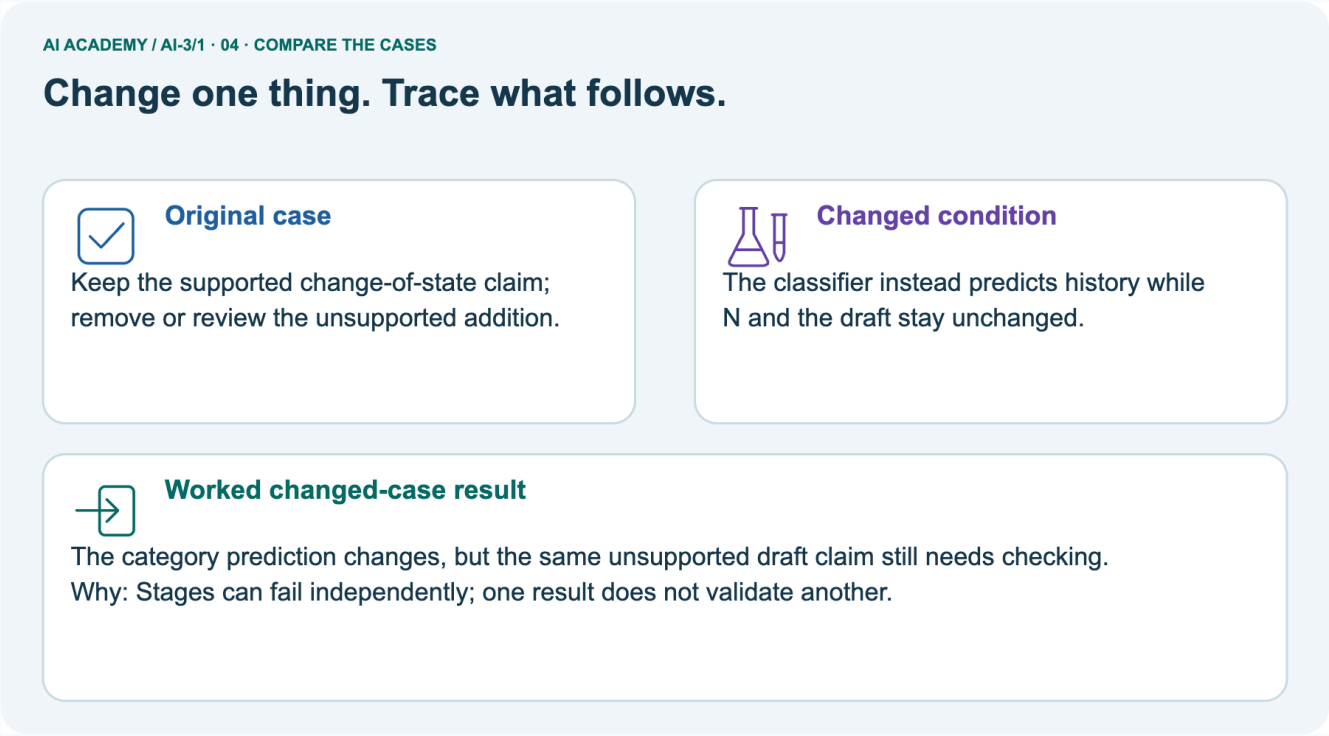
Keep the **evidence** roles separate. Ten corpus continuations | Count red 6, blue 3, green 1 | Distribution 0.6, 0.3, 0.1. Prefix: The marker is | Select a next token | An **output**, not an **observation**.

Current marker question | Check direct **evidence** | Colour claim needs support

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

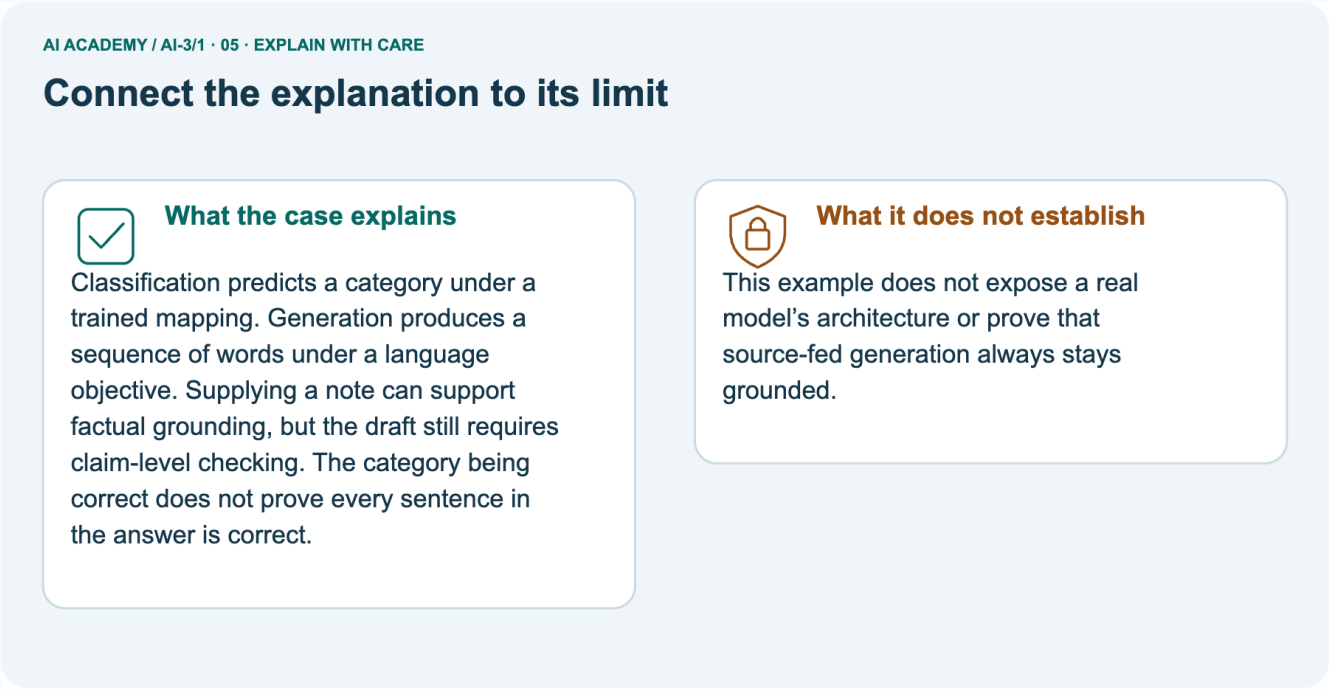
Original case: Keep the supported change-of-state claim; remove or review the unsupported addition. Changed condition: The classifier instead predicts history while N and the draft stay unchanged.

Expected result: The category **prediction** changes, but the same unsupported draft claim still needs checking. Reason: Stages can fail independently; one result does not validate another.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

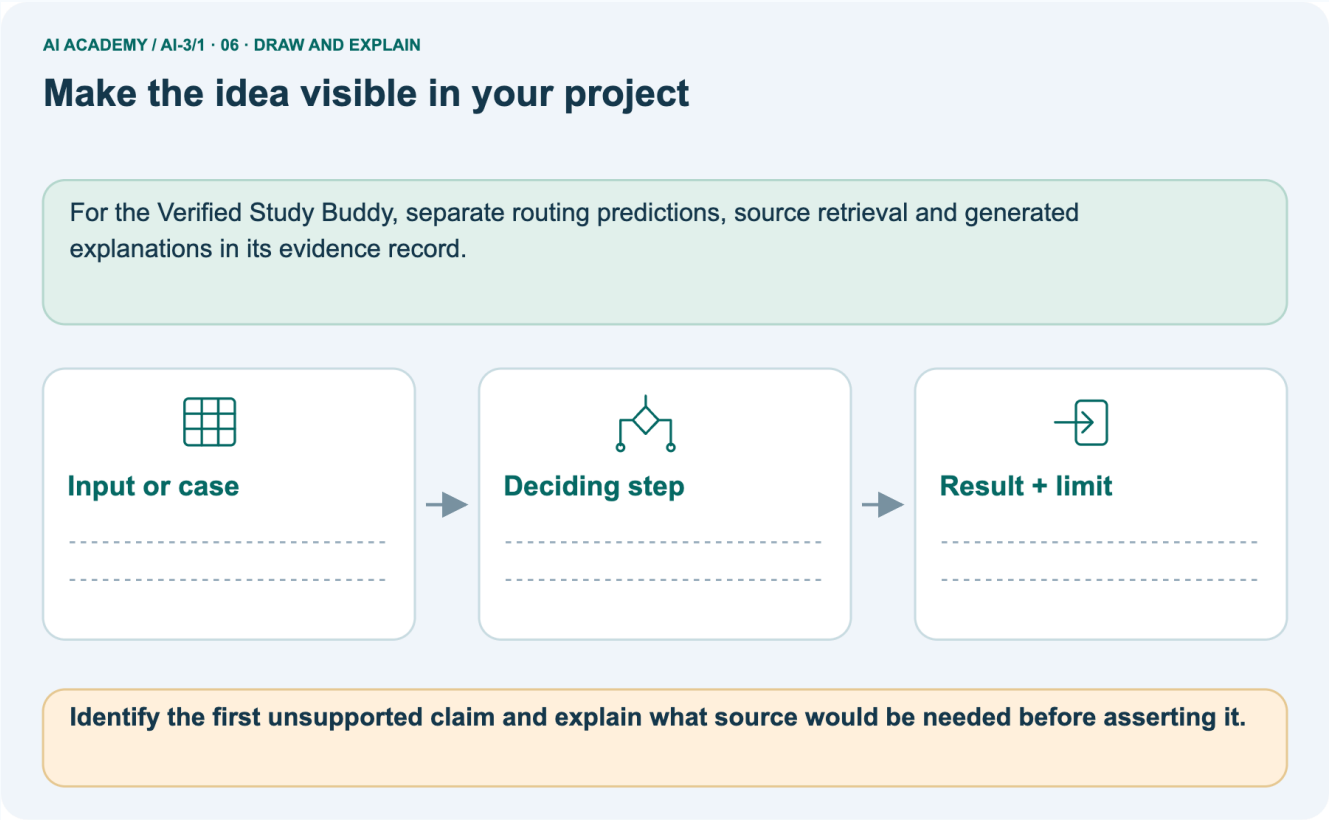
Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking.

The category being correct does not prove every sentence in the answer is correct. Limit: This example does not expose a real model's architecture or prove that source-fed generation always stays grounded.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Verified Study Buddy, separate routing **prediction**s, source retrieval and generated explanations in its **evidence** record. Draw your **input**, deciding step, result and limit.

Supported question: Identify the first unsupported claim and explain what source would be needed before asserting it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.